Graduate Admissions Outcomes: Analytics Report

Cleaned dataset size: 28,660 applicant entries (raw: 30,000). All values below are generated from the cleaned data.

1. Average GRE score of accepted PhD applicants

Mean GRE total = 327.28 (n = 312), using accepted PhD applicants with a GRE total in the 260-340 range.

2. Average GPA of accepted Master's applicants

Mean GPA = 3.72 (n = 3,793), using accepted Master's applicants with a GPA in the 0-4 range.

3. Shortcomings of inferring decision_date from status and term

- The status text gives only a month and day, so the year must be guessed from the term (e.g. Fall 2026). A Spring term is usually decided in the prior year, and this rule can be wrong near year boundaries.
- The term is the intended enrollment term, not the decision date, so the inferred year may be off by one for applicants who applied early or late.
- Status text is inconsistent (e.g. 'Interview', 'Wait listed', 'Other'), and some entries are intermediate updates, so inferred dates imply false precision.
- Entries are posted after decisions, so 11,640 rows have a negative days_to_decision; 11 rows have no decision_date at all.

4. Correlation analysis

GRE total vs GRE Verbal: Pearson r = 0.281, $p < 0.001$, n = 1,894. This is a weak positive linear association: applicants with higher total scores tend to have higher verbal scores, as expected because verbal is part of the total.

GPA vs GRE total: Pearson r = 0.044, p = 0.043, n = 2,116. This is a negligible positive association; GPA explains only a limited share of GRE variation.

Correlations use only rows with usable scores for both variables, so missing data may bias them. Correlation does not imply causation.

5. Do accepted and rejected applicants differ in GPA?

Accepted mean GPA = 3.762 (n = 7,010); rejected mean GPA = 3.781 (n = 7,978). Welch's t-test: t = -4.770, p < 0.001.

The difference (-0.019 GPA points, accepted minus rejected) is statistically significant but small in practice. Program, school and other applicant factors are not controlled for.

6. Chi-square test: Degree vs outcome

Contingency table (observed counts):

degree	Accepted	Interviewed	Rejected	Waitlisted
Master's	5047	325	1613	419
PhD	5417	1485	11932	2118
PsyD	105	75	93	20

chi-square = 4,437.66, df = 6, p < 0.001.

Degree type and outcome are statistically associated: outcome distributions differ across degree types. This is an association, not causation; program mix and applicant pool differences may contribute.

7. Remaining data-quality issues and limitations

- Missingness: GPA is missing for 39.2% of rows and GRE total for 91.8%; outcome is missing for 11 rows (statuses such as 'Other').

- Out-of-range values remain: 118 GPA values outside 0-4 and 967 GRE totals outside 260-340 (data entry errors or other scoring scales).

- Scores are self-reported and unverified; GPA scales may differ between applicants.

- Program and university names are free text and may be inconsistent or mis-split.

- The sample is voluntary self-reporting, so results do not generalize to all applicants.